

SUPPLY MANAGEMENT SYSTEM

Task Register

Addendum 29 - SCM Commercial Enablement

Business Partners | Sales Orders | Fulfillment | Invoicing | Customer Finance

Implementation Task Breakdown for Claude Code

Version 1.3 - Logistics-Aligned (Delivery Orders + IStockReservationService)

Property	Value
Document ID	SMS-TR-ADD-029
FSD Reference	SMS-FSD-ADD-029 v1.3
Version	1.0
Date	2026-09-19
Classification	Internal - Development
Total Tasks	68
Total Estimated Effort	30 Dev Days (240 Hours)
Architecture	.NET 8 / EF Core 8 / SQL Server 2022 / React 18
Status	Ready for Execution

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1. Executive Summary & Scope

This task register decomposes FSD Addendum 29 v1.3 (SCM Commercial Enablement) into granular, independently executable tasks. Each task maps directly to a section in the FSD, has explicit dependencies, estimated hours, and linked test case IDs.

Key alignment principle: This addendum reuses the existing logistics infrastructure (delivery_orders, IStockReservationService, pick/pack/goods-issue, carrier integration) rather than creating parallel structures. Tasks are structured accordingly — extending existing code rather than greenfield creation.

Scope Boundaries

- IN SCOPE: Business Partner migration, Sale Orders, Stock Reservation (SALES_ORDER source), Back-to-Back PO, Drop Ship, Self-Pickup, Sales Invoicing, Customer Ledger, Product Ledger, 10 Sales Reports, Email Intimation, PO-SO trace_id linkage
- IN SCOPE (Integration): Extend existing delivery_orders, IStockReservationService, GoodsIssueService, pick lists, pack API, carrier booking for sales fulfillment
- OUT OF SCOPE: Carrier onboarding (already built), Warehouse setup (already built), Supplier Scorecard (Addendum 23), Supplier Rate Cards (Addendum 28), Returns/Credit Notes (future addendum)

Technical Stack

Component	Technology
Backend	ASP.NET Core 8 / C# 12 / .NET 8 LTS
Database	SQL Server 2022 with EF Core 8 Code-First
ORM	EF Core 8 with dotnet ef migrations
Auth	ASP.NET Core Identity + JWT (org_id claim for multi-tenancy)
Background Jobs	Hangfire (hangfire schema in SQL Server)
PDF Generation	QuestPDF
Mapping	AutoMapper
Validation	FluentValidation
Events	MediatR
Frontend	React 18 + Vite + TypeScript
Multi-Tenancy	Shared DB, org_id discriminator, EF Core HasQueryFilter

2. Task Register Legend

Field	Description
Task ID	Unique identifier: A29-{phase}-{seq} (e.g., A29-P1-01)
Task Title	Concise action-oriented title
Description	What the task produces — entity, service, API, migration, or test
FSD Ref	Section reference in SMS-FSD-ADD-029 v1.3
Schema(s)	Database schema(s) affected
Depends On	Task IDs that must be completed first
Est. Hours	Estimated implementation hours (including unit tests)
Test IDs	Linked acceptance test case IDs from FSD Section 17

Task Type Color Coding

Type	Description	Naming Pattern
Migration	EF Core database migration	...migration / ...rename / ...add
Entity	C# entity class, EF configuration, DTO, AutoMapper profile	...entity / ...model
Service	Business logic service layer	...service / ...logic
API	Controller endpoints with FluentValidation	...api / ...endpoints
Integration	Extending existing service/entity for new source type	...extend / ...integrate
Frontend	React components, pages, API hooks	...ui / ...page / ...component
Test	Integration test, acceptance test	...test / ...verify
Hangfire	Background job definition and registration	...job / ...worker

Priority Levels

Priority	Description
P0 — Critical Path	Blocks multiple downstream tasks. Must be completed first.
P1 — High	Required for core functionality. Part of the main execution flow.
P2 — Medium	Important but has limited downstream dependencies.
P3 — Low	Enhancement, reporting, or polish. Can be deferred.

3. Phase 1 — Foundation (Business Partner Migration)

Priority: P0 — Critical Path | Estimated: 32 Hours (4 Days) | FSD Sections: 1.1–1.7

This phase renames Suppliers to BusinessPartners and adds carrier/customer/service-provider capabilities. Zero data loss. All existing FK references updated.

Task ID	Task Title	Description	FSD Ref	Schema(s)	Depends On	Est. Hours	Test IDs
A29-P1-01	Rename Suppliers → BusinessPartners (Migration)	EF Core migration: sp_rename [suppliers.Suppliers] to [suppliers.BusinessPartners]. Rename PK column supplier_id → partner_id. Update all FK references across procurement, inventory, finance schemas.	§1.1	suppliers, procurement, inventory, finance	—	4	TC-01
A29-P1-02	Add Partner Type Flags & Carrier/Service Fields (Migration)	Add columns: partner_type NVARCHAR(20), is_vendor BIT, is_customer BIT, is_carrier BIT, is_service_provider BIT, vehicle_types, service_categories, credit_limit, payment_terms_days. UPDATE existing rows SET partner_type='VENDOR', is_vendor=1.	§1.2, §1.3	suppliers	A29-P1-01	3	TC-01
A29-P1-03	BusinessPartner Entity & EF Configuration	C# entity: BusinessPartner with partner_type enum, four boolean flags. EF config: HasQueryFilter(p => p.OrgId == _orgId). AutoMapper profile: BusinessPartnerDto ↔ BusinessPartner. FluentValidation: partner_type must match flag combination.	§1.2, §1.3	suppliers	A29-P1-02	4	TC-01, TC-12
A29-P1-04	BusinessPartner Repository & Service	IBusinessPartnerRepository + BusinessPartnerService. Auto-compute partner_type from boolean flags on save. Filter methods: GetVendors(), GetCustomers(), GetCarriers(), GetServiceProviders(). Soft delete support.	§1.3–§1.5	suppliers	A29-P1-03	4	TC-01, TC-02, TC-03
A29-P1-05	BusinessPartner API Controller	PartnersController: GET/POST/PUT/DELETE /api/partners with filter params (type, is_vendor, is_customer, is_carrier, is_service_provider, active, search). Backward-compat aliases: /api/suppliers, /api/customers, /api/carriers, /api/service-providers.	§1.6, §1.7	suppliers	A29-P1-04	4	TC-01, TC-02, TC-03, TC-12
A29-P1-06	Update All Existing References (Supplier → Partner)	Rename SupplierLedger → PartnerLedger. Update all service/repository/controller references from Supplier to BusinessPartner. Update Scorecard (Add23) and RateCard (Add28) to filter by is_vendor=1. Verify all procurement flows unchanged.	§1.7	suppliers, procurement, finance	A29-P1-05	6	TC-01
A29-P1-07	BusinessPartner Unit & Integration Tests	Test: CRUD operations, partner_type auto-computation from flags, backward-compat /api/suppliers alias returns only vendors, FK integrity after rename, multi-tenant isolation.	§1.1–§1.7	suppliers	A29-P1-06	4	TC-01, TC-02, TC-03, TC-12
A29-P1-08	BusinessPartner Frontend Pages	React: Partners list page with type filter tabs (All/Vendors/Customers/Carriers/Service Providers). Partner create/edit form with dynamic fields based on type flags. Partner detail page with ledger tab.	§1.6	—	A29-P1-05	3	—

4. Phase 2 — Product & Pricing

Priority: P1 — High | Estimated: 16 Hours (2 Days) | FSD Sections: 2.1–2.4

Task ID	Task Title	Description	FSD Ref	Schema(s)	Depends On	Est. Hours	Test IDs
A29-P2-01	Add Default Supplier to ProductVariants (Migration)	Add columns to inventory.ProductVariants: default_supplier_id BIGINT NULL FK to BusinessPartners(is_vendor=1), lead_time_days INT NULL.	\$2.1	inventory	A29-P1-02	1	—
A29-P2-02	Create PricingRules Table (Migration)	Create inventory.PricingRules: pricing_rule_id, org_id, variant_id, partner_id (NULL=all), price_type (SELLING COST PROMOTIONAL CONTRACT), min_qty, max_qty, unit_price, currency_code, effective_from/to, is_active. HasQueryFilter on org_id.	\$2.2	inventory	A29-P1-02	2	—
A29-P2-03	PricingRule Entity, Service & Price Resolution Logic	IPricingService with ResolveSalePrice(variantId, partnerId, qty, date) implementing 5-priority resolution: CONTRACT → PROMOTIONAL → Partner SELLING → Default SELLING → Variant.selling_price. ResolvePurchasePrice: RateCard → LastPO → Variant.purchase_price.	\$2.3, \$2.4	inventory	A29-P2-02	5	—
A29-P2-04	PricingRules API Endpoints	PricingRulesController: CRUD /api/pricing-rules. GET /api/pricing-rules/resolve?variantId&partnerId&qty for price preview. Filter by variant, partner, type, active.	\$2.2–\$2.4	inventory	A29-P2-03	3	—
A29-P2-05	Pricing Unit Tests	Test 5-priority price resolution with edge cases: no rules, expired rules, overlapping qty ranges, partner-specific vs default, contract vs promotional priority.	\$2.3, \$2.4	inventory	A29-P2-03	3	—
A29-P2-06	Pricing Frontend (Admin)	React: Pricing rules CRUD page on Product Variant detail. Price type selector, date range pickers, quantity tiers. Price preview calculator.	\$2.2	—	A29-P2-04	2	—

5. Phase 3 — Sale Order Core

Priority: P0 — Critical Path | Estimated: 24 Hours (3 Days) | FSD Sections: 3.1–3.5, 4.1–4.2

Task ID	Task Title	Description	FSD Ref	Schemas	Dependencies	Est. Hours	Test IDs
A29-P3-01	Create SUPPLY_DEPT_ADMIN Role (Migration + Seed)	Create role in auth schema. Assign permissions: sale_order_config_read, sale_order_config_write. IT_ADMIN gets read-only. Seed role in migration.	\$3.1	auth	A29-P1-02	2	—
A29-P3-02	Create SaleOrderConfig Table (Migration)	Create demand.SaleOrderConfig: config_id, org_id (one per org), auto_po_enabled, supplier_selection_mode, auto_po_approval_mode, drop_ship_enabled, self_pickup_enabled, default_fulfillment_mode, reservation_ttl_hours, partial_fulfillment_allowed, email_intimation_enabled, intimation_department_id, intimation_cc_emails, shipment_required_default. HasQueryFilter on org_id.	\$3.2	demand	A29-P3-01	2	—
A29-P3-03	SaleOrderConfig Entity, Service & API	SaleOrderConfigService: GetConfig(orgId), UpdateConfig(orgId, dto). Audit trail on every change. API: GET/PUT /api/sale-order-config (SUPPLY_DEPT_ADMIN only for write). GET /api/sale-order-config/audit.	\$3.2–\$3.5	demand, auth	A29-P3-02	3	—
A29-P3-04	Create SaleOrders & SaleOrderLines Tables (Migration)	Create demand.SaleOrders: sale_order_id, org_id, so_number, partner_id (is_customer=1), order_date, expected_delivery_date, currency_code, subtotal, tax_amount, discount_amount, grand_total, status (DRAFT CONFIRMED PARTIALLY_FULFILLED FULFILLED INVOICED CLOSED CANCELLED), requires_shipment, delivery_mode (SHIP SELF_PICKUP), shipping_address_id FK to logistics.addresses, trace_id UNIQUEIDENTIFIER, is_deleted. Create demand.SaleOrderLines: so_line_id, sale_order_id, variant_id, quantity, unit_price, discount_percent, tax_percent, line_total, fulfilled_qty, invoiced_qty, fulfillment_mode (IN_STOCK BACK_TO_BACK DROP_SHIP SPLIT), available_qty_at_confirm, deficit_qty, linked_po_id, selected_supplier_id, margin, margin_percent, status. Indexes per §17.2.	\$4.1, \$4.2	demand	A29-P1-02, A29-P2-02	3	—
A29-P3-05	SaleOrder & SaleOrderLine Entities + EF Config	C# entities with enums (SaleOrderStatus, FulfillmentMode, DeliveryMode). EF config: HasQueryFilter(so => so.OrgId == _orgId), relationships to BusinessPartner, ProductVariant, logistics.Address. AutoMapper profiles. FluentValidation rules.	\$4.1, \$4.2	demand	A29-P3-04	4	TC-12
A29-P3-06	SaleOrder Service (CRUD + Document Numbering)	SaleOrderService: Create (DRAFT, auto-generate so_number via existing document_number_sequences pattern, generate trace_id = Guid.NewGuid()), Update (DRAFT only), GetById, GetList with filters. Price auto-resolution on line add via IPricingService. Line total calculation.	\$4.1, \$4.2, \$4.5	demand	A29-P3-05, A29-P2-03	5	TC-04, TC-05
A29-P3-07	SaleOrder API Controller	SaleOrdersController: GET/POST/PUT /api/sale-orders, GET /api/sale-orders/{id}, POST /api/sale-orders/{id}/confirm, POST /api/sale-orders/{id}/cancel, GET /api/sale-orders/{id}/timeline, GET /api/sale-orders/{id}/availability.	\$4.5	demand	A29-P3-06	3	TC-04, TC-05, TC-12
A29-P3-08	SaleOrder Config Frontend	React: Sale Order Configuration admin page (SUPPLY_DEPT_ADMIN role). Toggle controls for auto_po, drop_ship, self_pickup. Supplier selection mode dropdown. Approval mode selector. Audit log viewer.	\$3.2–\$3.5	—	A29-P3-03	2	—

6. Phase 4 — Availability Check, Reservation & Email Intimation

Priority: P0 — Critical Path | Estimated: 28 Hours (3.5 Days) | FSD Sections: 4.3–4.4, 5.1–5.3

Task ID	Task Title	Description	FSD Ref	Schem a(s)	Depe nds On	Est. Hou rs	Te st ID s
A2-9-P4-01	Extend IStockReservationService for SALES_ORDER	Add SALES_ORDER to ReservationSourceType enum. Extend Reserve() to accept SaleOrderLine source_id. Ensure FEFO multi-row allocation works for sales reservations. Add expiry support (expires_at from config TTL). No new table — uses existing inventory.stock_reservations.	\$4.4	inventory (shared)	A29-P3-05	4	TC-04, TC-05
A2-9-P4-02	Three-Scenario Availability Check Service	IAvailabilityCheckService.CheckAndReserve(saleOrderId): For each SO line, check available_qty. Scenario 1 (IN_STOCK): full reserve via IStockReservationService. Scenario 2 (SPLIT): partial reserve + deficit. Scenario 3 (BACK_TO_BACK): full deficit. Scenario 4 (DROP_SHIP): no stock impact. Set line fulfillment_mode, available_qty_at_confirm, deficit_qty.	\$4.3	demand, inventory	A29-P4-01, A29-P3-06	6	TC-04, TC-05, TC-06
A2-9-P4-03	SO Confirmation Orchestrator	Confirm SO transaction: validate DRAFT status, run availability check, reserve stock, create auto-POs for deficit (delegates to Phase 5), update SO status to CONFIRMED, enqueue email intimation via Hangfire. All in one transaction (except PO creation and email which are Hangfire jobs).	\$4.3, \$4.5	demand	A29-P4-02	4	TC-04, TC-05, TC-06
A2-9-P4-04	SO Cancellation Service	Cancel SO: validate status, release all ACTIVE reservations via IStockReservationService.Release(), cancel linked DRAFT POs, update SO status = CANCELLED, enqueue cancellation email.	\$4.5	demand, inventory	A29-P4-01	2	—
A2-9-P4-05	Reservation Expiry Hangfire Job	RecurringJob: scan stock_reservations WHERE source_type = SALES_ORDER AND status = ACTIVE AND expires_at < NOW(). Release expired reservations. Update SO line status. Send expiry warning email 24h before.	\$4.4, \$5.1	inventory, demand	A29-P4-01	2	—
A2-9-P4-06	Create SaleOrderIntimations Table (Migration)	Create demand.SaleOrderIntimations: intimation_id, org_id, sale_order_id, event_type (SO_CONFIRMED PO_CREATED DROP_SHIP RESERVED PO_APPROVED GRN_RECEIVED EXPIRING), recipients, subject, body_html, sent_at, status (SENT FAILED QUEUED), error_message, hangfire_job_id. HasQueryFilter on org_id.	\$5.3	demand	A29-P3-04	1	—
A2-9-P4-07	Email Intimation Service & Templates	ISaleOrderEmailService: SendConfirmation(sold), SendPOCreated(sold, pold), SendDropShip(sold), SendReserved(sold), SendPOApproved(pold), SendGRNReceived(grnId), SendExpiring(reservationId). HTML templates with line-by-line breakdown table. Hangfire BackgroundJob.Enqueue with 3x retry.	\$5.1, \$5.2, \$5.3	demand	A29-P4-06, A29-P4-03	6	TC-11
A2-9-P4-08	Availability & Email Unit Tests	Test: all 4 scenarios (in-stock, split, B2B, drop-ship), FEFO multi-batch reservation, expiry TTL, email template content validation, multi-tenant isolation of reservations.	\$4.3–\$4.4, \$5	demand, inventory	A29-P4-07	3	TC-04, TC-05, TC-11, TC-12

7. Phase 5 — Back-to-Back PO & trace_id / Timeline Linkage

Priority: P1 — High | Estimated: 32 Hours (4 Days) | FSD Sections: 3.3–3.4, 6.1–6.4, 13.1–13.7

Task ID	Task Title	Description	FSD Ref	Schema(s)	Dependencies	Est. Hours	Test IDs
A29-P5-01	Add PO Source & Linked Fields (Migration)	Add columns to procurement.PurchaseOrders: source NVARCHAR(20) DEFAULT 'MANUAL' (MANUAL BACK_TO_BACK DROP_SHIP MIR), linked_so_line_id BIGINT NULL, linked_so_id BIGINT NULL, customer_shipping_address_id BIGINT NULL FK to logistics.addresses, auto_generated_qty, auto_generated_price, auto_selected_supplier_id. Indexes per §17.2.	§6.3	procurement	A29-P3-04	2	—
A29-P5-02	Supplier Selection Service	ISupplierSelectionService with 3 modes: DEFAULT_SUPPLIER (variant.default_supplier_id fallback to MANUAL), BEST_MATCH (composite score = Scorecard*0.6 + PriceRank*0.4, tiebreak on lead_time_days), MANUAL (create task for supply team). Mode read from SaleOrderConfig.	§3.3	demand, suppliers	A29-P3-03, A29-P1-04	5	TC-06
A29-P5-03	Auto-PO Creation Service	IAutoPurchaseOrderService.CreateFromSODEficit(soLineId, supplierId, qty, price): Create PO with source=BACK_TO_BACK or DROP_SHIP, linked_so_id/linked_so_line_id, trace_id from SO. PO status per config: DRAFT_ONLY→DRAFT, REQUIRE_WORKFLOW→PENDING_APPROVAL, AUTO_SEND→APPROVED. Store auto_generated_qty/price/supplier for audit.	§6.1, §6.2	procurement, demand	A29-P5-01, A29-P5-02, A29-P2-03	6	TC-05, TC-06, TC-07
A29-P5-04	trace_id Propagation (SO → PO)	In AutoPO creation: copy saleOrder.trace_id to newPO.trace_id. Enqueue Hangfire timeline event PO_CREATED_FROM_SO via existing _timelineService.AppendEvent. Ensure org_id passed explicitly to Hangfire job.	§6.3.1, §13.3	workflow, procurement	A29-P5-03	3	TC-13
A29-P5-05	Drop Ship PO Flow	When fulfillment_mode=DROP_SHIP and config.drop_ship_enabled: Create PO with customer_shipping_address_id = SO.shipping_address_id, source=DROP_SHIP. No warehouse stock impact. PO vendor ships directly to customer address.	§4.3 Scenario 4, §14.1	procurement	A29-P5-03	2	—
A29-P5-06	GRN Auto-Reservation Link	Extend GRN approval service: after GRN for B2B PO (linked_so_line_id not null), auto-call IStockReservationService.Reserve(source_type=SALES_ORDER, source_id=soLineId) for received qty. Update SO line status OPEN→RESERVED. Enqueue GRN_FOR_SO_PO timeline event + email.	§6.4, §13.3	procurement, inventory, demand	A29-P5-04, A29-P4-01	4	TC-14
A29-P5-07	New Timeline Event Types	Add 7 event_type values to DocumentTimelines: SO_CREATED, SO_CONFIRMED, PO_CREATED_FROM_SO, GRN_FOR_SO_PO, SO_STOCK_RESERVED, SO_FULFILLED, SO_INVOICED. Extend existing TimelineService to emit these at each lifecycle point.	§13.3	workflow	A29-P5-04	3	TC-13, TC-14
A29-P5-08	Timeline Query API	Endpoints: GET /api/sale-orders/{id}/timeline, GET /api/purchase-orders/{id}/timeline (update existing), GET /api/timeline/{traceId}. Multi-tenancy enforced: org_id+trace_id unique index, org_id from JWT, Hangfire jobs pass org_id explicitly.	§13.5, §13.7	workflow	A29-P5-07	2	TC-13, TC-14, TC-15
A29-P5-09	B2B PO & Timeline Unit Tests	Test: 3 supplier selection modes, auto-PO creation with all 3 approval modes, trace_id propagation, GRN auto-reservation, timeline event chain (7+ events), non-blocking write (<2s confirm), multi-tenant timeline isolation.	§6, §13	procurement, workflow	A29-P5-08	3	TC-05, TC-06, TC-07, TC-13, TC-14, TC-15
A29-P5-10	Margin Calculation	On auto-PO creation: compute soLine.margin = selling_price - purchase_price, soLine.margin_percent. Stored on SaleOrderLines for reporting.	§14.2	demand	A29-P5-03	1	—
A29-P5-11	PO Modification Tracking	Track auto_generated_qty/price/supplier vs user-modified values on PO. Allow supply team to edit DRAFT B2B POs: update qty, supplier, price, add items, split PO. Audit log of changes.	§6.2	procurement	A29-P5-03	1	TC-07

Task ID	Task Title	Description	FSD Ref	Schema(s)	Dependencies	Est. Hours	Test IDs
	(Pre-Approval)						

8. Phase 6 — Fulfillment (Delivery Orders Integration)

Priority: P1 — High | Estimated: 24 Hours (3 Days) | FSD Sections: 7.1–7.9, 8.1–8.3

CRITICAL: This phase extends existing `logistics.delivery_orders` — it does NOT create new fulfillment tables.

Task ID	Task Title	Description	FSD Ref	Schema(s)	Dependencies	Est. Hours	Test IDs
A29-P6-01	Extend <code>delivery_orders</code> for <code>SALE_ORDER</code> Source (Migration)	Add columns to <code>logistics.delivery_orders</code> : <code>sale_order_id</code> BIGINT NULL FK, <code>delivery_mode</code> NVARCHAR(15) NULL, <code>pickup_person_name</code> , <code>pickup_person_id_type</code> , <code>pickup_person_id_number</code> , <code>pickup_person_id_authorization</code> . Add <code>SALE_ORDER</code> to <code>DeliverySourceType</code> enum. Add <code>so_line_id</code> BIGINT NULL to <code>delivery_order_lines</code> .	\$7.1	logistics	A29-P3-04	2	—
A29-P6-02	<code>DeliveryOrderService.CreateFromSaleOrder()</code>	Extend existing from-source creation pattern: create <code>delivery_order</code> with <code>from_source_type=SALE_ORDER</code> , <code>sale_order_id</code> , <code>delivery_mode</code> (SHIP/SELF_PICKUP), <code>ship_to_address_id</code> from <code>logistics.addresses</code> , <code>trace_id</code> from SO. Create <code>delivery_order_lines</code> with <code>so_line_id</code> mapping. Use existing <code>document_number_sequences</code> for DO number.	\$7.3	logistics, demand	A29-P6-01, A29-P3-06	4	TC-08, TC-09
A29-P6-03	Extend <code>GoodsIssueService</code> for <code>SALE_ORDER</code> Source	Add case <code>DeliverySourceType.SALE_ORDER</code> to <code>PostsGoodsIssue</code> : determine movement type (<code>SALES_SHIP</code> or <code>SALES_HANOVER</code> based on <code>delivery_mode</code>). Consume reservation via <code>IStockReservationService.Consume()</code> . Update <code>soLine.fulfilled_qty</code> . <code>Idempotent</code> (existing pattern).	\$7.5, \$8.3	logistics, inventory, demand	A29-P6-02, A29-P4-01	4	TC-08, TC-09
A29-P6-04	Self-Pickup Completion Endpoint	POST <code>/api/delivery-orders/{id}/pickup</code> : record <code>pickup_person_name</code> , <code>pickup_person_id_type/number</code> , <code>pickup_authorization</code> . Validate <code>delivery_mode=SELF_PICKUP</code> . Post goods issue (<code>SALES_HANOVER</code>). Advance status to <code>DELIVERED</code> . Generate gate pass PDF via <code>QuestPDF</code> (reuse existing packing list template).	\$8.2, \$8.3	logistics	A29-P6-03	3	TC-08
A29-P6-05	Sales Fulfillment Convenience APIs	POST <code>/api/sale-orders/{id}/create-delivery</code> : create DO from SO lines. GET <code>/api/sale-orders/{id}/deliveries</code> : list DOs for SO. All other operations use existing <code>/api/delivery-orders/{id}/*</code> endpoints (<code>pick</code> , <code>pack</code> , <code>goods-issue</code> , <code>ship</code> , <code>track</code> , <code>deliver</code>).	\$7.8	demand, logistics	A29-P6-02	2	TC-09
A29-P6-06	SO Status Update on Fulfillment Completion	When delivery order reaches <code>DELIVERED/COMPLETED</code> : update SO line <code>fulfilled_qty</code> , check if all lines fulfilled → SO status <code>PARTIALLY_FULFILLED</code> or <code>FULFILLED</code> . Emit <code>SO_FULFILLED</code> timeline event. Send fulfillment email.	\$7.6	demand, workflow	A29-P6-03, A29-P5-07	3	TC-09, TC-14
A29-P6-07	Fulfillment Integration Tests	Test: create DO from SO, <code>pick/pack/goods-issue/deliver</code> flow, self-pickup flow with gate pass, partial fulfillment (3 DOs from 1 SO), mixed delivery modes, SO status	\$7, \$8	logistics, demand, inventory	A29-P6-06	4	TC-08, TC-09, TC-12

Task ID	Task Title	Description	FSD Ref	Schema(s)	Dependencies	Est. Hours	Test IDs
		transitions, reservation consume on goods issue, multi-tenant isolation.					
A29-P6-08	Fulfillment Frontend	React: SO detail page 'Create Delivery' button → line/qty selector → , creates DO. Self-pickup form (person name, ID, authorization). Link to existing delivery order tracking pages. SO deliveries tab showing all DOs.	\$7.8 \$7.9	—	A29-P6-05	2	—

9. Phase 7 — Sales Invoicing & Customer Finance

Priority: P1 — High | Estimated: 32 Hours (4 Days) | FSD Sections: 9.1–9.5, 10.1

Task ID	Task Title	Description	FSD Schema Ref	Schema (s)	Dependencies	Est. Hours	Test IDs
A29-P7-01	Create SalesInvoices & SalesInvoiceLines Tables (Migration)	Create finance.SalesInvoices: invoice_id, org_id, invoice_number, sale_order_id, partner_id, invoice_date, due_date, subtotal, tax_amount, discount_amount, grand_total, amount_paid, balance_due, status (DRAFT ISSUED PARTIALLY_PAID PAID OVERDUE CANCELLED CREDIT_NOTE), currency_code, trace_id. Create finance.SalesInvoiceLines. Indexes per §17.2.	\$9.1, \$9.2	finance	A29-P3-04	2	—
A29-P7-02	Create CustomerPayments & PaymentAllocations Tables (Migration)	Create finance.CustomerPayments: payment_id, org_id, partner_id, payment_number, payment_date, amount, payment_method (CASH CHEQUE BANK_TRANSFER CARD ONLINE), cheque_number, bank_reference, currency_code, status (RECEIVED BOUNCED REVERSED). Create finance.PaymentAllocations: allocation_id, payment_id, invoice_id, allocated_amount.	\$9.3, \$9.4	finance	A29-P7-01	2	—
A29-P7-03	Create CustomerLedger Table (Migration)	Create finance.CustomerLedger: entry_id, org_id, partner_id, entry_date, entry_type (INVOICE PAYMENT CREDIT_NOTE DEBIT_NOTE ADVANCE REFUND OPENING_BAL), reference_type, reference_id, reference_number, debit_amount, credit_amount, running_balance, currency_code, narration. Mirrors existing SupplierLedger pattern. Indexes per §17.2.	\$10	finance	A29-P1-02	2	—
A29-P7-04	SalesInvoice Entity, Service & Business Rules	ISalesInvoiceService: CreateFromFulfillment(doid) — only from DELIVERED/COMPLETED DOs. Validate: invoiced_qty <= delivered_qty. Multiple invoices per SO (partial invoicing). On ISSUE: write CustomerLedger DEBIT entry in SAME transaction. Copy trace_id from SO. Emit SO_INVOICED timeline event.	\$9.5, \$13.3	finance, demand, workflow	A29-P7-01, A29-P6-06, A29-P5-07	6	TC-14
A29-P7-05	CustomerPayment Service & Auto-Allocation	ICustomerPaymentService: RecordPayment(partnerId, amount, method, details). Auto-allocate FIFO (oldest unpaid invoice first) with manual override. On RECEIVED: write CustomerLedger CREDIT entry in SAME transaction. Update invoice amount_paid/balance_due. Auto-set PARTIALLY_PAID when amount_paid > 0, PAID when balance_due = 0.	\$9.3, \$9.4, \$9.5	finance	A29-P7-02, A29-P7-04	5	—
A29-P7-06	CustomerLedger Service	ICustomerLedgerService: AppendEntry(partnerId, type, debit, credit, reference). Running balance calculation: prev_balance + debit - credit. Append-only pattern (same as SupplierLedger). GetLedger(partnerId, dateRange) with pagination.	\$10, \$10.1	finance	A29-P7-03	3	—
A29-P7-07	Invoice Overdue Hangfire Job	RecurringJob (daily): scan SalesInvoices WHERE status IN (ISSUED, PARTIALLY_PAID) AND due_date < TODAY. Set status = OVERDUE.	\$9.5	finance	A29-P7-04	1	—
A29-P7-08	Sales Invoice & Payment API Endpoints	SalesInvoicesController: CRUD, POST /issue, GET /pdf (QuestPDF). CustomerPaymentsController: POST, GET, POST /{id}/allocate. CustomerLedgerController: GET /api/partners/{id}/ledger (receivables).	\$9.1, \$9.5, \$10	finance	A29-P7-05, A29-P7-06	4	—
A29-P7-09	Invoice PDF Generation (QuestPDF)	Sales Invoice PDF: header (company, customer), line items table, totals, tax, payment terms, bank details. Gate Pass PDF (reuse from fulfillment). Both stored as document attachments.	\$9.1	finance	A29-P7-04	3	—
A29-P7-10	Invoicing & Payment Unit Tests	Test: invoice only against delivered qty, partial invoicing, payment auto-allocation FIFO, manual allocation, ledger running balance accuracy, overdue detection, multi-tenant isolation, bounced payment reversal.	\$9, \$10	finance	A29-P7-08	4	TC-12

10. Phase 8 — Product Ledger & Inventory Impact

Priority: P2 — Medium | Estimated: 16 Hours (2 Days) | FSD Sections: 11.1–11.4, 12.1–12.2

Task ID	Task Title	Description	FSD Ref	Schema(s)	Dependencies	Est. Hours	Test IDs
A29-P8-01	Create ProductLedger Table (Migration)	Create finance.ProductLedger: entry_id, org_id, variant_id, product_id (denormalized), entry_date, entry_type (PURCHASE SALE RETURN_IN RETURN_OUT ADJUSTMENT WRITE_OFF), reference_type, reference_id, reference_number, partner_id, quantity, unit_cost, total_cost, direction (IN OUT), running_qty, running_value, narration. Indexes per §17.2.	§11.1	finance	—	2	—
A29-P8-02	ProductLedger Entity & Service	IProductLedgerService: AppendEntry(variantId, type, qty, unitCost, direction, reference). WAC calculation: on IN → new_value = prev_value + (qty*cost), new_WAC = new_value/new_qty. On OUT → COGS = current WAC. Written in SAME transaction as GRN approval or Invoice issuance.	§11.2, §11.3	finance	A29-P8-01	5	TC-10
A29-P8-03	Integrate ProductLedger with GRN & Sales Invoice	On GRN approval: write PURCHASE entry (IN, unit_cost = PO line price). On Sales Invoice issuance: write SALE entry (OUT, unit_cost = WAC, not selling price). Both in same transaction as business action.	§11.2	finance, procurement, demand	A29-P8-02, A29-P7-04	3	TC-10
A29-P8-04	Add Sales Movement Types to Existing MovementType Enum	Add SALES_SHIP, SALES_HANOVER, SALES_RETURN, DROP_SHIP_VIRTUAL to existing MovementType enum used by GoodsIssueService. Note: stock_balances reserved_qty/available_qty already exist from IStockReservationService (G4 decision).	§12.1, §12.2	inventory	A29-P6-03	1	—
A29-P8-05	ProductLedger API Endpoints	GET /api/product-ledger/{variantId}: paginated entries. GET /api/product-ledger/{variantId}/summary: total purchased, sold, current stock value, WAC. GET /api/reports/product-profitability: revenue vs COGS by product.	§11.4	finance	A29-P8-02	2	TC-10
A29-P8-06	Product Ledger Unit Tests	Test: WAC calculation across multiple GRNs at different prices, COGS on sale, running_qty/running_value accuracy, zero-stock edge case, multi-tenant isolation.	§11	finance	A29-P8-03	3	TC-10

11. Phase 9 — Reports & Frontend

Priority: P3 — Low | Estimated: 24 Hours (3 Days) | FSD Sections: 15, Frontend

Task ID	Task Title	Description	FSD Ref	Schema(s)	Depends On	Est. Hours	Test IDs
A29-P9-01	R1: Sales Order Register Report	GET /api/reports/sales/order-register: all SOs with status, totals, customer, delivery mode. Filters: date range, status, customer, delivery_mode. PDF (QuestPDF) + Excel export.	\$15 R1	reports	A29-P3-07	2	—
A29-P9-02	R2: Customer Ledger Report	GET /api/reports/sales/customer-ledger: receivables ledger per customer. Filters: customer, date range.	\$15 R2	reports	A29-P7-06	2	—
A29-P9-03	R3: Aging Receivables Report	GET /api/reports/sales/aging-receivables: outstanding invoices by age bucket (0-30, 31-60, 61-90, 90+ days). Filters: as-of date, customer.	\$15 R3	reports	A29-P7-04	2	—
A29-P9-04	R4–R6: Sales by Product, Customer, Fulfillment Status	Three reports: revenue/qty by product, revenue/count/avg by customer, open fulfillments by status/warehouse/delivery mode. All with PDF+Excel export.	\$15 R4–R6	reports	A29-P7-04, A29-P6-06	4	—
A29-P9-05	R7–R8: Margin Analysis & Sales vs Purchase	Margin Analysis: selling price vs cost, margin% by product/customer/SO. Sales vs Purchase Comparison: revenue vs COGS, gross margin by period.	\$15 R7–R8	reports	A29-P8-02, A29-P5-10	3	—
A29-P9-06	R9–R10: Product Ledger & Profitability Reports	Product Ledger Report: all movements per product with running qty/value. Product Profitability: revenue - COGS per product, ranked by margin.	\$15 R9–R10	reports	A29-P8-05	3	—
A29-P9-07	Sale Order Frontend Pages	React: SO list page with status filters. SO create/edit form with line items, customer selector, delivery mode toggle. SO detail page with tabs: Lines, Deliveries, Invoices, Payments, Timeline. Confirm/Cancel actions.	\$4.5, \$7.8	—	A29-P3-07, A29-P6-05	4	—
A29-P9-08	Sales Invoice & Payment Frontend	React: Invoice list/detail pages. Create invoice from delivery. Invoice PDF viewer. Payment recording form with method selector. Payment allocation UI. Customer ledger view.	\$9, \$10	—	A29-P7-08	3	—
A29-P9-09	Reports Dashboard Frontend	React: Sales Reports dashboard with all 10 reports. Filter controls, PDF download, Excel export. Chart visualizations for margin and aging reports.	\$15	—	A29-P9-06	1	—

12. Phase 10 — Integration Testing & Acceptance

Priority: P1 — High | Estimated: 12 Hours (1.5 Days) | FSD Section: 17

Task ID	Task Title	Description	FSD Ref	Schema(s)	Depends On	Est. Hours	Test IDs
A29-P10-01	TC-01: Vendor Backward Compatibility	Verify: GET /api/suppliers returns only is_vendor=1. Create PO with existing vendor. All procurement flows unchanged after rename.	\$17 TC-01	suppliers, procurement	A29-P1-07	1	TC-01
A29-P10-02	TC-04/05: SO In-Stock & SPLIT Scenarios	Test full flow: SO with 50 available → reserve all (TC-04). SO with 60/100 available → 60 reserved + PO for 40 (TC-05). Verify email content, reservation records, PO creation.	\$17 TC-04, TC-05	demand, inventory, procurement	A29-P4-08, A29-P5-09	2	TC-04, TC-05
A29-P10-03	TC-08: Self-Pickup Full Flow	Test: SO with SELF_PICKUP → DO created → pick → pack → goods issue (SALES_HANDOVER) → record pickup person → DELIVERED. Verify gate pass PDF, stock deduction, no consignment.	\$17 TC-08	logistics, demand, inventory	A29-P6-07	1	TC-08
A29-P10-04	TC-09: Partial Fulfillment (3 DOs)	Test: SO for 100 units → DO-1 (40 shipped) → DO-2 (30 shipped) → DO-3 (30 self-pickup). Verify SO status transitions: CONFIRMED→PARTIALLY_FULFILLED→FULFILLED. Mixed delivery modes.	\$17 TC-09	logistics, demand	A29-P6-07	1	TC-09
A29-P10-05	TC-10: Product Ledger WAC Accuracy	Test: GRN 100@\$10 → Sell 30@\$15. Verify PURCHASE entry (IN), SALE entry (OUT, COGS=WAC=\$10). Running_qty=70, running_value=\$700.	\$17 TC-10	finance	A29-P8-06	1	TC-10
A29-P10-06	TC-12: Multi-Tenant Isolation	Test: Create customer in Org A. Login as Org B. Verify: cannot access Org A customer, cannot reference in SO, timeline events isolated, reservations isolated, invoices isolated.	\$17 TC-12	all	A29-P7-10	2	TC-12
A29-P10-07	TC-13/14: trace_id Full Chain Test	Test: SO confirmed (SPLIT) → auto-PO → PO approved → GRN → auto-reserve → 2 DOs → invoices. GET /api/timeline/{traceld}: verify all 7+ events chronologically. Verify trace_id identical across all documents.	\$17 TC-13, TC-14	workflow, all	A29-P5-09, A29-P7-10	2	TC-13, TC-14
A29-P10-08	TC-15: Timeline Non-Blocking Write	Test: Confirm SO that triggers auto-PO. Measure: SO confirmation response < 2 seconds. Timeline event appears within 5 seconds (async via Hangfire). Verify business transaction not slowed.	\$17 TC-15	workflow	A29-P5-08	1	TC-15
A29-P10-09	End-to-End Smoke Test	Full cycle: Create customer → Create SO → Confirm (SPLIT) → Approve auto-PO → Receive GRN → Create DO → Pick → Pack → Goods Issue → Ship → Deliver → Invoice → Payment → Verify ledgers + timeline + reports.	\$17	all	A29-P10-07	1	All

13. Dependency Graph

Critical path is highlighted. Tasks within a phase can be parallelized where dependencies allow.

Phase 1: Business Partner Migration (P0)

A29-P1-01 → A29-P1-02 → A29-P1-03 → A29-P1-04 → A29-P1-05 → A29-P1-06 → A29-P1-07
↳ A29-P1-08 (parallel)

Phase 2: Product & Pricing (P1) – starts after A29-P1-02

A29-P2-01 $\neg$
A29-P2-02 $\neg$ $\rightarrow$ A29-P2-03 $\rightarrow$ A29-P2-04 $\rightarrow$ A29-P2-05
 $\hookrightarrow$ A29-P2-06 (parallel)

Phase 3: Sale Order Core (P0) – starts after P1 + P2

A29-P3-01 → A29-P3-02 → A29-P3-03 → A29-P3-08 (parallel)
A29-P3-04 → A29-P3-05 → A29-P3-06 → A29-P3-07

Phase 4: Availability & Email (P0) – starts after P3

A29-P4-01 → A29-P4-02 → A29-P4-03 → A29-P4-07
 A29-P4-04 (parallel after P4-01)
 A29-P4-05 (parallel after P4-01)
 A29-P4-06 → A29-P4-07 → A29-P4-08

Phase 5: B2B P0 & Timeline (P1) – starts after P3 + P4

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A29-P5-01 → A29-P5-02 → A29-P5-03 → A29-P5-04 → A29-P5-07 → A29-P5-08 → A29-P5-09
                                     ↳ A29-P5-05 (parallel)
                                     ↳ A29-P5-10 (parallel)
                                     ↳ A29-P5-11 (parallel)
A29-P5-06 (after P5-04 + P4-01)

```

Phase 6: Fulfillment / Delivery (P1) – starts after P3 + P4

A29-P6-01 → A29-P6-02 → A29-P6-03 → A29-P6-04
 ↳ A29-P6-05 → A29-P6-08 (parallel)
 A29-P6-06 (after P6-03 + P5-07)
 A29-P6-07 (after P6-06)

Phase 7: Invoicing & Finance (P1) – starts after P6

A29-P7-01 → A29-P7-04 → A29-P7-09 (parallel)
A29-P7-02 → A29-P7-05
A29-P7-03 → A29-P7-06
A29-P7-07 (after P7-04)
A29-P7-08 (after P7-05 + P7-06) → A29-P7-10

Phase 8: Product Ledger (P2) – starts after P7

A29-P8-01 → A29-P8-02 → A29-P8-03 → A29-P8-06
A29-P8-04 (after P6-03) A29-P8-05 (after P8-02)

Phase 9: Reports & Frontend (P3) – starts progressively

A29-P9-01..06 (after respective data phases)
A29-P9-07..09 (after respective API phases)

Phase 10: Integration Testing (P1) – starts after all phases
A29-P10-01..09 (after all dependent phases)

14. Effort Summary by Phase

Phase	Title	Tasks	Hours	Days	Priority
Phase 1	Business Partner Migration	8	32	4.0	P0
Phase 2	Product & Pricing	6	16	2.0	P1
Phase 3	Sale Order Core	8	24	3.0	P0
Phase 4	Availability, Reservation & Email	8	28	3.5	P0
Phase 5	B2B PO & trace_id Linkage	11	32	4.0	P1
Phase 6	Fulfillment (Delivery Orders)	8	24	3.0	P1
Phase 7	Sales Invoicing & Finance	10	32	4.0	P1
Phase 8	Product Ledger & Inventory	6	16	2.0	P2
Phase 9	Reports & Frontend	9	24	3.0	P3
Phase 10	Integration Testing	9	12	1.5	P1
	TOTAL	83	240	30.0	

Effort by Work Type

Work Type	Approximate Hours	% of Total
Migrations (EF Core)	20	8%
Entities & EF Config	25	10%
Services & Business Logic	65	27%
API Controllers	25	10%
Integration (Extend Existing)	20	8%
Frontend (React)	17	7%
Unit & Integration Tests	35	15%
Reports & PDFs	20	8%
Hangfire Jobs	5	2%
End-to-End Testing	8	3%

15. Risk Register

ID	Risk	Impact	Likelihood	Mitigation
R1	BusinessPartner rename breaks existing FK references	High	Medium	Careful sp_rename order. Run all existing procurement tests after migration. Backward-compat aliases.
R2	StockReservationService FEFO allocation race condition under concurrent SO confirmations	High	Low	Existing RowVersion concurrency on stock_reservations. Unit test with parallel reservations.
R3	delivery_orders state machine assumes PO/SRO source — SALE_ORDER source may hit unexpected validations	Medium	Medium	Review all state transition guards for source-type assumptions. Add SALE_ORDER to all relevant switch statements.
R4	GoodsIssueService PostsGoodsIssue does not handle SALES_HANDOVER movement type	Medium	Low	Extend switch with explicit case. Unit test both SALES_SHIP and SALES_HANDOVER paths.
R5	Auto-PO creation in SO confirmation transaction causes long-running transaction	High	Medium	PO creation via Hangfire BackgroundJob.Enqueue (non-blocking). Only reservation is in main transaction.
R6	Product Ledger WAC calculation drift from rounding	Low	Medium	Use DECIMAL(18,4) everywhere. Test with multi-GRN scenarios. Periodic reconciliation report.
R7	Email intimation Hangfire job failures not surfaced to users	Medium	Low	3x retry with exponential backoff. FAILED status logged in SaleOrderIntimations. Admin alert on repeated failures.
R8	Existing 1531 logistics tests break after delivery_orders schema changes	High	Low	All new columns are NULL-able additions. Existing tests should pass unchanged. Run full logistics test suite after migration.

16. Migration Script Execution Order

Execute in sequence. Each migration is independently revertible with dotnet ef database update <previous>.

#	Migration Name	Phase	Schema(s)	Description
M01	RenameSupplierToBusinessPartner	P1	suppliers	Rename table + PK column, update FKs
M02	AddPartnerTypeFlagsAndCarrierFields	P1	suppliers	Add type flags, carrier/service fields
M03	AddDefaultSupplierToVariants	P2	inventory	Add default_supplier_id to ProductVariants
M04	CreatePricingRules	P2	inventory	New PricingRules table
M05	CreateSupplyDeptAdminRole	P3	auth	New role + permissions seed
M06	CreateSaleOrderConfig	P3	demand	New SaleOrderConfig table
M07	CreateSaleOrderTables	P3	demand	SaleOrders + SaleOrderLines tables
M08	AddSalesOrderToReservationSourceType	P4	inventory	Extend ReservationSourceType enum
M09	CreateSaleOrderIntimations	P4	demand	New SaleOrderIntimations table
M10	AddPOSourceAndLinkedFields	P5	procurement	Add source, linked_so_id, etc. to POs
M11	AddSOTimelineEvents	P5	workflow	Register 7 new event types
M12	ExtendDeliveryOrdersForSaleOrderSource	P6	logistics	Add sale_order_id, delivery_mode, pickup fields to DOs
M13	CreateSalesInvoiceTables	P7	finance	SalesInvoices + SalesInvoiceLines
M14	CreateCustomerPaymentTables	P7	finance	CustomerPayments + PaymentAllocations
M15	CreateCustomerLedger	P7	finance	New CustomerLedger table
M16	CreateProductLedger	P8	finance	New ProductLedger table

Total: 16 migrations across 6 schemas (suppliers, inventory, auth, demand, procurement, workflow, logistics, finance).

16.1 Rollback Strategy

- Each migration is separate and revertible: dotnet ef database update <previous-migration>
- M01 (BusinessPartner rename): reversible by renaming table back to Suppliers
- M02–M04: nullable column additions, safe to drop
- M05–M09: new tables only, safe to drop without affecting existing data
- M10: nullable column additions to PurchaseOrders, safe to drop
- M11: additive to DocumentTimelines event types (no destructive impact)
- M12: nullable column additions to delivery_orders, safe to drop
- M13–M16: new tables only, safe to drop
- SALES_ORDER removal from ReservationSourceType requires clearing sales reservations first
- All 1531 existing logistics tests must pass after M12